Geocentric epicycle expansions of Keplerian orbits: explicit coefficients, error bounds, and frequency analysis

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The idea that Ptolemaic epicycles are "just Fourier series" is old but usually stated loosely. We make it precise for Newtonian motion. A bound Kepler orbit is an exact Fourier series in the mean anomaly with Bessel-function coefficients, and each term is a pair of counter-rotating circles; the geocentric position of a body is the sum of two such families, one for the body and one for the Earth, and is almost-periodic rather than periodic. We give the explicit three-dimensional transform from orbital elements to epicycle radii, planes and rates, with rigorous truncation bounds. Truncations reach 10^{-6} to 10^{-9} AU with about 6 to 10 harmonics per orbit for Mars and Ceres but need hundreds at eccentricity 0.89; a circular two-term approximation of Mars (one deferent and one epicycle, with no eccentric or equant, and so not Ptolemy's actual model) reproduces all retrograde loops but with longitude errors up to 33°. Numerical frequency analysis (NAFF) recovers the analytic terms from Keplerian data with relative frequency error $\sim 2 \times 10^{-10}$. For an N-body Ceres trajectory over 200 years, compared with six-element Kepler fits on the same heliocentric data and with the same integrated Earth subtracted, the interior-window error decreases with the number of lines K (monotonically for $K \geq 10$) to 2×10^{-5} AU at $K = 200$, but the full-window error is edge-limited at a few 10^{-2} AU and is comparable to the fitted Kepler orbit. A tighter-tolerance rerun of the integration changes the reference by at most 10^{-8} AU. The representation is exact for Kepler motion and on analytic invariant tori, is an empirical finite-window approximation for other perturbed motion, and does not exist for unbound orbits. The order certified by the truncation bound grows as $(1 - e)^{-3/2}$ near $e = 1$, but the actual minimal order at fixed tolerance tends to a finite limit. We also provide an interactive three-dimensional viewer and the code to reproduce every table and figure. The conceptual correspondence is not new; our contribution is the explicit quantitative geocentric construction.

I. INTRODUCTION

Ptolemy's *Almagest* [1] predicted planetary positions well enough to serve as the working astronomy of the Mediterranean and Islamic worlds for well over a thousand years, using circles carried by other circles [2, 3], with motion that is uniform about the circle's centre or, for the equant, about an offset point (Sec. II). That a scheme so alien to modern dynamics could be so accurate invites a modern explanation. A popular one is that epicycles are "just Fourier series": a sum of rotating vectors can trace any closed curve, so of course it fits the sky [4–7]. The statement is true in spirit but, as usually made, it is loose in three ways. First, the motion of a planet seen from Earth involves at least two generally incommensurate frequencies, the mean motions of the planet and of the Earth, so the geocentric path is not periodic and has no Fourier series in the strict sense; the appropriate setting is that of Bohr's almost-periodic functions [8]. Second, Ptolemy's actual models use eccentrics and the equant and are not harmonic expansions [2, 9]. Third, if epicycles can fit anything, the representation is empirically empty [10], unless the radii and rates are fixed by something other than the data.

In this paper we ask a precise version of the question. Given a body on a Newtonian orbit, what is the epicycle system, with radii and angular rates determined by the orbital elements rather than fitted, that reproduces its geocentric position, how many circles does it need for a prescribed accuracy, and for which motions does the answer remain valid? The answer has two parts. For unperturbed Kepler motion the heliocentric position is an exact Fourier series in the mean anomaly with Bessel-function coefficients [11–13]. Subtracting the Earth's position gives a sum of two such series, one for the body and one for the Earth, which is an epicycle system in which every radius and rate is a function of $(a, e, i, \omega, \Omega, M_0)$ of the two orbits. For perturbed motion the representation is exact only on analytic invariant tori (whose existence for a given trajectory is not guaranteed); for other trajectories it is an empirical finite-window approximation, whose frequencies and amplitudes can be recovered numerically by Laskar's frequency analysis [14–16].

What is, and is not, claimed. The correspondence between epicycles and Fourier or quasi-periodic expansions is not new, and we do not claim to have discovered it. It is stated explicitly by Gallavotti [17]; the Bessel expansion of Kepler motion goes back to Lagrange and Bessel [11, 18]; and modern planetary theories are, in practice, enormous sums of uniform circular motions [19, 20]. Section III surveys this literature. What we found no source to do together, after a search described there, is the following, and these are the contributions of this paper.

1. An explicit, closed-form, three-dimensional geocentric epicycle expansion of an arbitrary Keplerian

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body, with Bessel (Kapteyn) amplitudes, built from the decomposition of each ellipse into two counter-rotating circles, with inclination handled exactly as uniformly rotating vectors in tilted planes.

2. Truncation-error bounds for this expansion, and the resulting number of epicycles needed for a given accuracy as a function of eccentricity, for both the body and the Earth terms.

3. A numerical frequency-analysis (NAFF) study. On unperturbed Keplerian geocentric trajectories (Mars, Eros), NAFF recovers the analytic Bessel epicycle terms (for the ten leading Mars terms over 200 years, with 10-day sampling and 24 lines, relative frequency error 2×10^{-10} and relative radius error 9×10^{-8}). On an N-body-perturbed Ceres trajectory (Sun, Earth, Jupiter and Saturn; 200 years), the NAFF reconstruction error is compared with that of least-squares-fitted six-element Kepler orbits, fitted to the same heliocentric data with matched weights and with the same integrated Earth subtracted from every reconstruction; NAFF is clearly better than the fitted Kepler orbit only for $K \gtrsim 50$ lines and only in the interior of the window, where edge effects dominate the full window.

4. A precise statement of validity: the representation is exact for Kepler motion and on analytic invariant tori, an empirical finite-window approximation for other perturbed motion (including chaotic motion), and does not apply to unbound orbits.

In addition we provide an open-source interactive three-dimensional visualization of the resulting epicycle chains as a pedagogical tool. The claim of novelty is about our search, not a proof of absence.

Why this is not curve fitting. In the construction below, every radius and angular rate is determined by the orbital elements of the body and of the Earth. Epicycles thus become a derived coordinate representation of Newtonian motion rather than a fitting model, which is our answer to the objection that an epicycle system explains nothing because it can reproduce anything.

Outline. Section II gives brief historical background and Sec. III reviews prior work and positions our contribution. Section IV derives the expansion and its truncation bounds, and Sec. V describes the numerical methods (Kepler propagation, N-body integration, and NAFF). Section VI reports the results, Sec. VII describes the visualization, and Sec. VIII discusses the domain of validity before Sec. IX concludes. Bessel-function identities used in the derivation are collected in the Appendix.

II. HISTORICAL BACKGROUND

The geometry we use is ancient. Apollonius is credited with showing that the motion of a body on an eccentric circle is equivalent to motion on an epicycle carried by a concentric deferent [2], and Hipparchus is the traditional source of the quantitative use of these devices for the Sun and the Moon [2, 3]. In the *Almagest*, Ptolemy combined deferent, epicycle and eccentric for every planet, and introduced the equant, a point offset from the deferent's centre, opposite the Earth, about which the epicycle's centre, moving along the deferent, sweeps out equal angles in equal times [1, 2]. The equant is not a harmonic device; Evans showed that it reproduces Kepler's second law to first order in the eccentricity [9], and the later history of the division of the Martian eccentricity can be read as a sequence of approximations to Kepler motion [21].

Copernicus changed the origin of the frame but not the circle paradigm: his models still need epicycles, with epicyclets replacing the equant [22]. The step that makes the geocentric motion a *difference* of two heliocentric motions is already visible for circular orbits, where the Earth's orbit becomes the epicycle of a superior planet [3, 23]. Kepler's ellipses then replaced the circles [24], yet an ellipse is not an escape from circular motion: expanding Kepler's equation in a Fourier series in the mean anomaly leads to Bessel-function coefficients, a result usually dated to Lagrange and Bessel in the early nineteenth century [11]. In this reading the ellipse is an infinite sum of uniform circular motions, with coefficients that decay geometrically (Sec. IV B). The connection between epicycles and quasi-periodic expansions was drawn explicitly by Gallavotti [17], and Sec. III places our construction relative to that literature. We cite these historical works through secondary sources throughout and have not consulted the primary texts of Bessel or Apollonius directly.

III. PRIOR WORK

The idea that epicycles and Fourier-type expansions are the same thing, and the individual ingredients of our construction, have a long history. We organize it by the ingredient it supplies. The survey is the result of a literature search whose limits we state plainly: we cannot exclude that relevant older or non-English sources escaped it, and several primary sources (Bessel's 1824 memoir, Schiaparelli's 1874 work) were not consulted directly and are cited through secondary accounts or not at all.

Epicycles as a universal representation. Hanson [4] argued that epicyclic astronomy is mathematically powerful enough to reproduce essentially any observed path, and that its decline was not due to weak mathematics. This is the classical statement of the "epicycles can draw anything" thesis, repeated in popular treatments [5–7] and in a short approximation-theory note [25]. It is a kinematic statement and is not derived from dynamics. Its philosophical consequence, that a system able to fit any curve is hard to refute, is discussed by Carman [10, 26]. Manin [27] and Boccaletti [24] place the con-

tinuity of circle-composition ideas in broader historical context.

Epicycles as quasi-periodic motion. The closest conceptual precedent is Gallavotti [17]. He rewrites a quasi-periodic Fourier series as a deferent carrying nested epicycles ordered by amplitude, expands the Kepler ellipse of Mars in powers of the eccentricity to obtain a deferent and two epicycles at first order and more at higher order, and argues, through Kolmogorov's theorem, that motions near integrable ones are again epicyclic, with approximation only over finite times in general. Our work does not supersede this; as far as we can tell, it is planar, does not write the geocentric difference of two orbits explicitly, and does not use Bessel coefficients or treat three dimensions. The adjacent mathematical framework is Bohr's theory of almost-periodic functions [8], which is where we place the geocentric expansion. Scholz [23] shows that moving from heliocentric circular orbits to an Earth-based observer leads to epicyclic constructions, which is our geocentric transform restricted to circular orbits.

The Bessel–Fourier expansion of Kepler motion. That a Kepler orbit has an exact Fourier expansion in the mean anomaly with Bessel-function coefficients is classical. It originates with Lagrange and Bessel [11, 28] and its convergence theory is that of Kapteyn series [12, 18]; expansions of the radius and anomalies are standard textbook material [13, 29], surveyed by Colwell [30], with recent work on summation and convergence [31, 32]. A point worth keeping separate is that the Laplace limit $e \approx 0.6627$ governs the power-series solution in e, not the Fourier series in the mean anomaly, which converges for all $e < 1$ [28, 30]. Our expansion rests on these results; what we add is their assembly into a three-dimensional geocentric epicycle chain with explicit truncation bounds (Sec. IV).

Ptolemaic models as truncations of Kepler motion. Evans showed that the equant reproduces Kepler's second law to first order in eccentricity [9] and traced the successive historical approximations to Kepler motion for Mars [21]. Rushkin [33] analyzed the accuracy attainable with Ptolemy's eccentricity-division rule, finding errors of second order in e. Carman and Recio [34] give the most careful modern quantitative comparison of Ptolemaic and Keplerian longitude models, with optimized parameters. We do not claim to give the first Ptolemy-versus-Kepler comparison. The standard historical sources for the models themselves are [1–3, 22].

Newtonian dynamics to epicycles. Several works derive epicycles from Newtonian orbits. Peralta, Calles and Yépez [35] integrate Newton's equations and apply an FFT to the resulting time series, showing that the Fourier terms generate the deferent and epicycles of the Copernican model; this is the most direct precedent for the numerical half of our programme, though heliocentric and planar, without perturbations, and without the analytic Bessel transform. Sugon, Bragais and McNamara [36] obtain Copernican epicycles from the linearized equation of motion about a circular orbit, to first order and in a heliocentric frame. Pucacco [37] expands the Kepler problem in a co-rotating frame, obtaining at first order a circle plus a second circle at twice the frequency, which he calls an epicyclic expansion, and extends it with normal forms to perturbed cases. Paillusson and Booth [38] compute geocentric transforms of heliocentric trajectories to discuss the underdetermination of world systems, without a Fourier decomposition. An unreviewed software repository [39] reconstructs planar geocentric trajectories from Keplerian elements as sums of rotating circles using FFT peaks and least squares, which shows that the geocentric numerical idea is in circulation, so our novelty cannot rest on the idea alone.

Frequency analysis and ephemeris series. Operationally, planetary positions are already represented as sums of uniform circular motions. VSOP87 [19] expresses heliocentric coordinates as long sums of Poisson terms, and each term with no secular factor is a uniform circular motion; a geocentric version is obtained by differencing two such theories. Kudryavtsev [20] obtained analytical series for DE431 by spectral fitting of a numerical ephemeris. Laskar's numerical frequency analysis [14–16, 40] recovers quasi-periodic decompositions with an accuracy far better than a plain FFT, with convergence estimates in terms of the data length [41], and also marks the limits of the epicycle picture: the motion of the solar system is chaotic, and over long times frequencies drift so that a fixed set of epicycles ceases to represent the solution [14, 42]. We use NAFF as our numerical tool and reuse these limits in our statement of validity.

Position of the present work. To summarize, concept and framing are due to Gallavotti [17]; the Bessel expansion to Lagrange, Bessel and Kapteyn; the FFT demonstration to Peralta et al. [35]; first-order analytic derivations to Sugon et al. [36] and Pucacco [37]; and the practical representation to VSOP87 and Laskar-type frequency analysis. We found no source that combines (i) the explicit closed-form three-dimensional geocentric expansion with ellipse-as-two-circles structure, (ii) rigorous truncation bounds, (iii) a frequency-analysis study that checks NAFF against the analytic expansion on unperturbed geocentric trajectories and, for a perturbed (N-body) geocentric Ceres trajectory, compares its reconstruction error with fitted-Kepler baselines, with the limits of that comparison stated, and (iv) a precise statement of the domain of validity. That combination, together with the visualization of Sec. VII, is what this paper contributes, as an expository synthesis with new quantitative results rather than as a new discovery.

IV. THEORY

This section derives the geocentric epicycle expansion of a Newtonian orbit. We keep the classical background (Subsecs. IV A and IV B) separate from the material we believe to be our contribution: the assembly of vector

terms into counter-rotating circles in three dimensions (Subsec. IV C), the geocentric chain (Subsec. IV D), the explicit truncation bounds (Subsec. IV F), and an explicit delimitation of where the representation holds for perturbed motion (Subsec. IV G), whose substance is classical. The identification of the leading terms with the Ptolemaic deferent, epicycle and equant (Subsec. IV E) is known and is included for completeness. Throughout, a and e are the semi-major axis and eccentricity of a Keplerian orbit, $s \equiv \sqrt{1-e^2}$, $M = n(t-t_p)$ is the mean anomaly, E the eccentric anomaly with $M = E - e\sin E$, and $z = x + \mathrm{i}y$ is the position in the perifocal plane (x toward periapsis, origin at the focus). Orientation is given by the rotation $\mathsf{R} = \mathsf{R}_z(\Omega)\mathsf{R}_x(i)\mathsf{R}_z(\omega)$.

A. Kepler motion as a Fourier series in the mean anomaly

Classical background [11–13, 29]; re-derived here to fix notation. In the perifocal frame,

$$x = a(\cos E - e), \qquad y = as\sin E. \tag{1}$$

For $0 \le e < 1$ the map $M \mapsto E$ is a smooth increasing bijection of the circle, so $z(M)$ is real-analytic and 2π-periodic in M and has a uniformly convergent Fourier series. Writing the trigonometric functions of E as exponentials,

$$\frac{z}{a} = -e + \frac{1+s}{2}\,e^{\mathrm{i}E} + \frac{1-s}{2}\,e^{-\mathrm{i}E}, \tag{2}$$

a prograde circle of radius $(1+s)/2$, a retrograde one of radius $(1-s)/2$, and an offset, all in the *eccentric* anomaly. What remains is to re-expand $e^{\pm\mathrm{i}E}$ in the uniform variable M.

Lemma 1 (Bessel [11], Watson [12], ch. 17)**.** *For integers $m \neq 0$ and $k \neq 0$,*

$$\frac{1}{2\pi}\int_0^{2\pi} e^{\mathrm{i}mE}e^{-\mathrm{i}kM}\,dM = \frac{m}{k}\,J_{k-m}(ke). \tag{3}$$

Proof. Integrate by parts in M (the boundary term vanishes by periodicity), change variable to E using $M = E - e\sin E$, and use Bessel's integral $J_n(x) = \frac{1}{2\pi}\int_0^{2\pi} e^{\mathrm{i}(n\tau - x\sin\tau)}d\tau$ (see Appendix A). □

Applying Lemma 1 to Eq. (2) with $m = \pm 1$ gives the following.

Proposition 1 (Fourier–Bessel series of Kepler motion)**.** *$z = a\sum_{k\in\mathbb{Z}} c_k e^{\mathrm{i}kM}$ with all c_k real, where for $k \ge 1$*

$$\begin{aligned} c_k &= \frac{(1+s)J_{k-1}(ke) - (1-s)J_{k+1}(ke)}{2k} \\ &= \frac{1}{k}\Big[\frac{1+s}{e}J_k(ke) - J_{k+1}(ke)\Big], \end{aligned} \tag{4}$$

$$\begin{aligned} c_{-k} &= \frac{(1-s)J_{k-1}(ke) - (1+s)J_{k+1}(ke)}{2k} \\ &= \frac{1}{k}\Big[\frac{1-s}{e}J_k(ke) - J_{k+1}(ke)\Big], \end{aligned} \tag{5}$$

and $c_0 = -\frac{3}{2}e$. Equivalently, in real form,

$$\begin{aligned} \frac{x}{a} &= -\frac{3e}{2} + \sum_{k\ge1}\frac{2}{k}J_k'(ke)\cos kM, \\ \frac{y}{a} &= \frac{2s}{e}\sum_{k\ge1}\frac{1}{k}J_k(ke)\sin kM, \end{aligned} \tag{6}$$

that is, $c_k + c_{-k} = \frac{2}{k}J_k'(ke)$ and $c_k - c_{-k} = \frac{2s}{ke}J_k(ke)$.

The second forms in Eqs. (4) and (5) follow from the recurrences $J_{k-1} + J_{k+1} = 2kJ_k/x$ and $J_{k-1} - J_{k+1} = 2J_k'$ and the relation $J_k'(x) = kJ_k(x)/x - J_{k+1}(x)$ (Appendix A). The offset follows from averaging Eq. (2) over M with $dM = (1 - e\cos E)\,dE$: $\langle\cos E\rangle_M = -e/2$ and $\langle\sin E\rangle_M = 0$.

The limit $e \to 0$. The factors s/e and $(1\pm s)/e$ in Eqs. (4)–(6) are singular at $e = 0$ term by term, but the coefficients are not: since $J_k(ke) = O(e^k)$, one has $J_k(ke)/e \to \frac{1}{2}\delta_{k1}$, so all c_k stay finite and tend to $c_1 = 1$, $c_{-1} = 0$, and $c_k = 0$ for all other k. The case $e = 0$ is therefore the single circle $z = ae^{\mathrm{i}M}$. For small e,

$$\begin{aligned} c_0 = -\tfrac{3}{2}e, \quad c_1 = 1 - \tfrac{1}{2}e^2 + \dots, \quad c_2 = \tfrac{1}{2}e + \dots, \\ c_{-1} = \tfrac{1}{8}e^2 + \dots, \quad c_{-2} = \tfrac{1}{24}e^3 + \dots, \end{aligned} \tag{7}$$

so $z/a \approx -\frac{3}{2}e + e^{\mathrm{i}M} + \frac{e}{2}e^{2\mathrm{i}M}$ to first order in e. (Appendix A gives the general small-e behaviour.)

One term is one epicycle. With $M = n(t-t_p)$, the term $ac_ke^{\mathrm{i}kM}$ is a uniform circular motion of radius $a|c_k|$, angular rate kn (counter-clockwise for $k > 0$, clockwise for $k < 0$) and phase $\arg c_k \in \{0, \pi\}$. The $k = 0$ term is a fixed displacement $-\frac{3}{2}ea$ along the apsidal line of the centre of all the circles.

B. Convergence and the Kapteyn bound

Classical [12, 18], with the corollaries for c_k spelled out. Kapteyn's inequality states that for $0 < \zeta \le 1$ and $\nu \ge 0$, $|J_\nu(\nu\zeta)| \le \big(\zeta e^{\sqrt{1-\zeta^2}}/(1+\sqrt{1-\zeta^2})\big)^\nu$ (Appendix A). With $\zeta = e$:

$$|J_k(ke)| \le \rho(e)^k, \qquad \rho(e) = \frac{e\,e^s}{1+s} < 1 \quad (0 \le e < 1). \tag{8}$$

The function ρ increases from $\rho(0) = 0$ to $\rho(1) = 1$, and near $e = 1$, with $\delta = 1 - e$, $\ln\rho \approx -(2\delta)^{3/2}/3$.

Proposition 2 (Coefficient decay and convergence)**.** *For $k \ge 1$ and $0 < e < 1$,*

$$\begin{aligned} |c_k| \le \frac{\rho^k}{k}\Big[\frac{1+s}{e} + \rho\Big], \quad |c_{-k}| \le \frac{\rho^k}{k}\Big[\frac{1-s}{e} + \rho\Big], \\ |c_k| + |c_{-k}| \le \frac{2\rho^k}{k}\Big[\frac{1}{e} + \rho\Big]. \end{aligned} \tag{9}$$

Hence $\sum_k |c_k| < \infty$ and the series of Prop. 1 converges absolutely and uniformly in M for every $0 \le e < 1$.

Proof. Because ρ is increasing, $|J_{k+1}(ke)| = |J_{k+1}((k+1)\frac{ke}{k+1})| \le \rho(\frac{ke}{k+1})^{k+1} \le \rho(e)^{k+1}$. Insert this and Eq. (8) in the derivative-free forms of Eqs. (4), (5). Uniform convergence follows by the Weierstrass M-test since $\rho < 1$. □

The Laplace limit $e \approx 0.6627$ is sometimes quoted as a bound on the convergence of such expansions. It is the radius of convergence of the Lagrange power series of $E(M;e)$ in powers of e at fixed M: continued to complex e, $E(M;e)$ has branch points, located where $1 - e\cos E = 0$ at complex E, the nearest of which lies at $|e| = 0.6627$, and a power series cannot converge beyond the nearest singularity. The value 0.6627 is a worst-case statement over the mean anomaly: it is the largest e for which the series converges for *every* M, not the radius of convergence at each fixed M (at $M = 0$ and $M = \pi$, $E = M$ identically and the series is trivial). It does not limit the Fourier series in M at fixed real e, whose coefficients decay geometrically at rate $\rho(e) < 1$ for *all* $e < 1$ (Prop. 2). The two expansions order the terms differently (powers of e versus harmonics of M); re-expanding $c_k(e)$ in e recovers the Lagrange series, which then inherits the Laplace limit [28, 30].

C. Three dimensions: each vector term as two counter-rotating circles

Let $\mathbf{P}, \mathbf{Q}, \mathbf{N}$ be the columns of R: the periapsis direction, the in-plane quadrature direction and the orbit normal, an orthonormal right-handed triad ($\mathbf{P}\times\mathbf{Q} = \mathbf{N}$). Then $\mathbf{r} = x\mathbf{P} + y\mathbf{Q} = \mathrm{Re}\big[z(\mathbf{P} - \mathrm{i}\mathbf{Q})\big]$ and, using that the c_k are real,

$$\mathbf{r}(M) = a\,c_0\,\mathbf{P} + \sum_{k\ge1} \mathrm{Re}\big(\mathbf{A}_k e^{\mathrm{i}kM}\big),$$
$$\mathbf{A}_k = a\big[c_k(\mathbf{P} - \mathrm{i}\mathbf{Q}) + c_{-k}(\mathbf{P} + \mathrm{i}\mathbf{Q})\big] \in \mathbb{C}^3. \qquad (10)$$

Equivalently $\mathbf{A}_k = a[\alpha_k\mathbf{P} - \mathrm{i}\beta_k\mathbf{Q}]$ with $\alpha_k = c_k + c_{-k}$ and $\beta_k = c_k - c_{-k}$, so each harmonic is an ellipse centred at the origin, with semi-axes $a|\alpha_k|$ along $\mathbf{P}$ and $a|\beta_k|$ along $\mathbf{Q}$, traversed at rate kn in the plane orthogonal to $\mathbf{N}$.

Proposition 3 (Two counter-rotating circles per harmonic)**.** *With $\phi_{\pm k} = \arg c_{\pm k} \in \{0, \pi\}$,*

$$\mathrm{Re}\big(\mathbf{A}_k e^{\mathrm{i}kM}\big) = \underbrace{a|c_k|[\cos(kM + \phi_k)\mathbf{P} + \sin(kM + \phi_k)\mathbf{Q}]}_{\textit{prograde, } +kn}$$
$$+ \underbrace{a|c_{-k}|[\cos(kM - \phi_{-k})\mathbf{P} - \sin(kM - \phi_{-k})\mathbf{Q}]}_{\textit{retrograde, } -kn}. \qquad (11)$$

The ellipse has semi-major axis $a(|c_k| + |c_{-k}|)$ and semi-minor axis $a\,\big||c_k| - |c_{-k}|\big|$, and is a circle iff $c_{-k} = 0$.

Proof. Expand $\mathrm{Re}[c_k e^{\mathrm{i}kM}(\mathbf{P} - \mathrm{i}\mathbf{Q})]$ and $\mathrm{Re}[c_{-k}e^{\mathrm{i}kM}(\mathbf{P} + \mathrm{i}\mathbf{Q})]$ using $c_{\pm k} = \pm|c_{\pm k}|$ (i.e. $e^{\mathrm{i}\phi} = \pm 1$); the sum of a circle of radius r_+ at rate $+\omega$ and one of radius r_- at rate $-\omega$ is an ellipse with semi-axes $r_+ + r_-$ and $|r_+ - r_-|$. □

The Kepler case has orthogonal ellipse axes along $\mathbf{P}$ and $\mathbf{Q}$. For a general complex vector amplitude $\mathbf{A} \in \mathbb{C}^3$ at rate ω (as returned by frequency analysis, or for any harmonic in Eq. (18)), $\mathrm{Re}(\mathbf{A}e^{\mathrm{i}\omega t}) = \mathsf{B}\,(\cos\omega t, \sin\omega t)^{\mathsf{T}}$ with the real 3×2 matrix $\mathsf{B} = [\mathrm{Re}\,\mathbf{A}, -\mathrm{Im}\,\mathbf{A}]$. The curve is an ellipse centred at the origin. Let $\mathsf{B} = \sum_{j=1,2}\sigma_j\mathbf{u}_j\mathbf{v}_j^{\mathsf{T}}$ be its singular value decomposition, with $\sigma_1 \ge \sigma_2 \ge 0$, and choose the signs of $(\mathbf{u}_2, \mathbf{v}_2)$ so that $\det[\mathbf{v}_1\ \mathbf{v}_2] = +1$ (this leaves B unchanged). Then, for a phase ϕ fixed by $\mathbf{v}_1$, $\mathrm{Re}(\mathbf{A}e^{\mathrm{i}\omega t}) = \sigma_1\mathbf{u}_1\cos(\omega t + \phi) + \sigma_2\mathbf{u}_2\sin(\omega t + \phi)$, which is the sum of a prograde circle of radius $(\sigma_1 + \sigma_2)/2$ and a retrograde circle of radius $(\sigma_1 - \sigma_2)/2$ in the plane spanned by $\mathbf{u}_1, \mathbf{u}_2$ (senses taken about $\mathbf{u}_1 \times \mathbf{u}_2$). A linearly polarised harmonic ($\sigma_2 = 0$) is two circles of equal radius $\sigma_1/2$, and a circle has $\sigma_1 = \sigma_2$. Proposition 3 is the special case of Kepler motion, where the ellipse axes are known in advance (along $\mathbf{P}$ and $\mathbf{Q}$) and no decomposition is needed. This is the conversion used for the NAFF lines in Sec. V.

For a single Kepler body all circles lie in the same orbital plane, so Prop. 3 is the planar statement of the previous subsections written in the basis $\mathbf{P}, \mathbf{Q}$. Inclination matters once two orbital planes are combined (next subsection): the body's circles lie in the plane with normal $\mathbf{N}_b$ and the Earth's in the plane with normal $\mathbf{N}_\oplus$, which gives ecliptic latitude with no additional machinery. In this form the construction requires no small parameter in the inclination.

D. The geocentric chain

Let $\boldsymbol{r}_{\mathrm{geo}} = \mathbf{r}_b - \mathbf{r}_\oplus$ with $M_b = n_b(t - t_b)$ and $M_\oplus = n_\oplus(t - t_\oplus)$; the perihelion times are absorbed into the phases of the amplitudes. By Eq. (10),

$$\boldsymbol{r}_{\mathrm{geo}}(t) = \underbrace{a_b c_0^{(b)}\mathbf{P}_b - a_\oplus c_0^{(\oplus)}\mathbf{P}_\oplus}_{\textit{constant offset}} + \sum_{k\ge1}\mathrm{Re}\big(\mathbf{A}_k^{(b)} e^{\mathrm{i}kn_b t}\big)$$
$$- \sum_{j\ge1}\mathrm{Re}\big(\mathbf{A}_j^{(\oplus)} e^{\mathrm{i}jn_\oplus t}\big). \qquad (12)$$

Proposition 4 (Geocentric epicycle chain)**.** *(i) The Cartesian spectrum of $\boldsymbol{r}_{\mathrm{geo}}$ is* contained in *the union of the two harmonic ladders $\{kn_b\} \cup \{jn_\oplus\}$; there are no combination frequencies $kn_b \pm jn_\oplus$ in the Cartesian vector (they appear only in nonlinear functions of it such as ecliptic longitude, distance or angular rate). (ii) Eq. (12) is a uniformly and absolutely convergent trigonometric series with two base frequencies, i.e. a Bohr almost-periodic function [8] with frequency module $\mathbb{Z}n_b + \mathbb{Z}n_\oplus$. If $n_b/n_\oplus$ is irrational, $\boldsymbol{r}_{\mathrm{geo}}$ is not periodic and has no Fourier series in a single period. (iii) If $n_b/n_\oplus = p/q$ with p, q coprime positive integers, then $n_b = p\nu$ and $n_\oplus = q\nu$ with $\nu = n_b/p = n_\oplus/q$, all frequencies are integer multiples of ν, and $\boldsymbol{r}_{\mathrm{geo}}$ is periodic with period*

$$T = \frac{2\pi}{\nu} = \frac{2\pi p}{n_b} = \frac{2\pi q}{n_\oplus}, \qquad (13)$$

which is the minimal period whenever the $k = 1$ terms of both bodies survive in Eq. (12) (then $p\nu$ and $q\nu$ are both present and $\gcd(p, q) = 1$). In general the minimal period is set by the frequencies that actually survive the subtraction.

Proof. (i) is linearity of the difference. Equality can fail: a circular orbit ($e = 0$) has no harmonics above $k = 1$, and if the two orbits coincide the difference vanishes identically, so coefficients vanish or cancel and the spectrum can be a proper subset. (ii) follows from Eq. (9), which makes both ladders absolutely summable. (iii) is the arithmetic above; note that $2\pi q/n_b = (q/p)T$ is a period only when $p = 1$. □

For Mars, $n_b/n_\oplus \approx 0.5317$ (mean elements), and the synodic period is a derived quantity of the lattice, not an exact period. Generic (irrational) frequency ratios give motion on a two-torus [17].

E. The leading Ptolemaic elements

Keep only $k = 1$ for both bodies, neglect e ($c_1 \to 1$, $c_0, c_{-1} \to 0$) and work in the ecliptic plane with complex notation:

$$z_{\text{geo}} \approx a_b e^{\mathrm{i}\theta_b} - a_\oplus e^{\mathrm{i}\theta_\oplus}, \qquad \theta = nt + \text{const.} \tag{14}$$

For an *outer* planet the deferent is the body's $k = 1$ circle (radius a_b, period $2\pi/n_b$) and the epicycle is the Earth's $k = 1$ term (radius $a_\oplus$, period one year); the epicycle-to-deferent ratio is $a_\oplus/a_b$, e.g. 0.656 for Mars. The vector from the epicycle centre to the planet is $-\mathbf{r}_\oplus$, hence always parallel to the Earth-to-Sun direction: the classical rule that the epicycle radius stays parallel to the mean-Sun line [2, 3]. For an *inner* planet the roles swap: the deferent is the Sun's geocentric circle $-a_\oplus e^{\mathrm{i}\theta_\oplus}$ and the epicycle is the planet's own circle, with ratio $a_b/a_\oplus$ (e.g. 0.723 for Venus). Which circle is called the deferent is the commutativity of the sum (14), the algebraic content of Apollonius's equivalence [2, 3]. (We do not compare with Ptolemy's own numerical parameters.)

Including the first-order eccentricity terms, $z/a \approx -\frac{3}{2}e + e^{\mathrm{i}M} + \frac{e}{2}e^{2\mathrm{i}M}$ by Eq. (7). For the equant, place the central body at $O = 0$, the deferent centre at $C = -ea$ and the equant point at $Q = -2ea$ (so C bisects OQ). A point on the deferent circle of radius a about C, whose direction from Q advances uniformly at angle M, is $P = Q + \varrho e^{\mathrm{i}M}$ with $\varrho \approx a + \mathrm{Re}(\bar{d}e^{\mathrm{i}M})$, $d = C - Q = ea$, so

$$\begin{aligned} P - O &\approx -2ea + ae^{\mathrm{i}M} + \tfrac{ea}{2}\big(e^{2\mathrm{i}M} + 1\big) \\ &= a\Big[-\tfrac{3e}{2} + e^{\mathrm{i}M} + \tfrac{e}{2}e^{2\mathrm{i}M}\Big] + O(e^2). \end{aligned} \tag{15}$$

Proposition 5 (Ptolemaic models at leading order)**.** *The circular deferent–epicycle model is the $k = 1$ truncation of Eq. (12) at $e = 0$. The eccentric-plus-equant model with bisected eccentricity reproduces the $k = 0, 1, 2$ terms of the Kepler series to first order in e, so that its error is $O(e^2)$: the missing pieces are $c_{-1} = O(e^2)$, $c_{-2} = O(e^3)$ and the $O(e^2)$ corrections to c_1, c_2.*

That the equant and bisected eccentricity capture Kepler's equation to $O(e)$ is known [9, 21, 33, 34]. The equant model is a different parametrisation of the same three leading terms and is not itself a finite harmonic-lattice expansion.

F. Truncation error

Let $z_N = a\sum_{|k|\le N} c_k e^{\mathrm{i}kM}$ ($2N$ circles plus the offset).

Theorem 1 (Uniform truncation bound)**.** *For $0 < e < 1$,*

$$\begin{aligned} \sup_t |z - z_N| &\le a \sum_{k>N} \big(|c_k| + |c_{-k}|\big) \\ &\le \varepsilon_N(a, e) \equiv a\,\frac{2(1/e + \rho)\,\rho^{N+1}}{(N+1)(1-\rho)}. \end{aligned} \tag{16}$$

Since $z(M)$ is exactly 2π-periodic in M, the supremum over M equals that over all t: the bound holds uniformly for all time for an unperturbed bound orbit. For the geocentric chain, keeping $|k| \le N_b$ for the body and $|k| \le N_\oplus$ for the Earth,

$$\begin{aligned} \sup_t |\boldsymbol{r}_{\text{geo}} - \boldsymbol{r}^{(N)}_{\text{geo}}| \\ \le \varepsilon_{N_b}(a_b, e_b) + \varepsilon_{N_\oplus}(a_\oplus, e_\oplus). \end{aligned} \tag{17}$$

Proof sketch. The first inequality is the triangle inequality. The second applies Eq. (9) and $\sum_{k>N}\rho^k/k \le \frac{1}{N+1}\sum_{k>N}\rho^k = \rho^{N+1}/[(N+1)(1-\rho)]$. For Eq. (17), R is an isometry, so the Cartesian errors of the body and the Earth add by the triangle inequality. □

As $e \to 0$ the bound is $O(\rho^{N+1}/e) \to 0$ for $N \ge 1$ (with $\varepsilon_0 = a$ at $e = 0$, the dropped circle). The number of harmonics *certified* by the bound for tolerance tol (position error $\sup_M |z - z_N|/a <$ tol, i.e. tolerance relative to a) scales as $N_{\text{cert}} \sim \ln(1/\text{tol})/\ln(1/\rho)$ up to logarithmic factors, and near $e \to 1$ as $N_{\text{cert}} \propto (1 - e)^{-3/2}$, because $\ln(1/\rho) \approx (2\delta)^{3/2}/3$. This is the order that Eq. (16) *guarantees* as $\rho \to 1$; it is a statement about the geometric decay scale of the bound, not about the smallest order that actually meets the tolerance. The two differ qualitatively as $e \to 1$. At $e = 1$ the coefficients tend to $c_{\pm k} \to J'_k(k)/k$ (Eqs. (4) and (5) with $s = 0$), and the uniform Bessel expansions [43] give $J'_k(k) \simeq Ck^{-2/3}$ with $C = 3^{1/6}\Gamma(2/3)/(2^{1/3}\pi) \approx 0.41085$. The coefficients thus decay only algebraically, $c_{\pm k} \simeq Ck^{-5/3}$, and the limiting position tail is $\sum_{|k|>N}|c_k| \simeq 3CN^{-2/3}$. At fixed tolerance relative to a the actual minimal order therefore tends to a *finite* limit, $N_{\text{actual}}(\text{tol}) \to (3C/\text{tol})^{3/2}$ as

TABLE I. Harmonics N needed for the sampled maximum of $|z-z_N|/a$ over a fine M grid to fall below tol (observed), and from the tail-sum bound (16).

e	$\rho(e)$	obs.		bound	
		10^{-6}	10^{-12}	10^{-6}	10^{-12}
0.0167 (Earth)	0.0227	4	7	4	8
0.0934 (Mars)	0.1267	6	12	7	13
0.25	0.3345	10	22	12	24
0.6	0.7418	33	75	43	87
0.89	0.9644	230	575	350	712

$e \to 1$, rather than diverging as $(1-e)^{-3/2}$ (the limit itself grows as $\mathrm{tol}^{-3/2}$). The asymptotic estimates are those of the limiting coefficients; the numbers in Table II are evaluated directly from the exact coefficients. Table I compares the bound with the smallest N actually needed (the sampled maximum of $|z-z_N|/a$ on a 3001-point grid on $[0,\pi]$, the error being symmetric under $M \to -M$); the bound overestimates N by at most a factor of about 1.5 in this range, and was checked to hold for every N from 0 to 800 (to 6000 for $e = 0.89$).

Table II extends the comparison towards $e \to 1$. It uses $\sup_M |z - z_N|/a$ with the tolerance held fixed relative to a. The coefficients $c_{\pm k}$ were evaluated numerically up to $k_{\max}$ between 10^6 and 10^7 and found non-negative at every k examined (the asymptotics $c_{\pm k} \simeq Ck^{-5/3} > 0$ suggest this persists beyond; we do not prove it). Given non-negativity, every term of the tail is largest at $M = 0$ (periapsis), so the sampled maximum and the tail at $M = 0$ coincide, and in the cases computed the sampled maximum was attained at $M \approx 0$ (to within the sampling resolution near $M = 0$). We use this identification to compute N_{actual} from the coefficient sum. It is a numerical observation with an asymptotic argument, not a theorem. Entries marked with an asterisk are extrapolated with the limiting law $3CN^{-2/3}$, because the tail had not reached the tolerance at $k_{\max}$; they are approximate estimates, not counts. At tolerance $10^{-2}a$ the minimal order is 1058 at $e = 0.999$, 1327 at 0.9999 and 1366 at 0.999999, approaching the limit ≈ 1367 at $e = 1$, whereas N_{cert} grows by a factor of about 32 per decade of $1-e$ and exceeds 4.7×10^6 already at $e = 0.9999$. At $10^{-6}a$ the same saturation is expected at $\approx 1.4 \times 10^9$, far beyond any practical chain. Truncation to a given absolute error at high e is thus costly, but the $(1-e)^{-3/2}$ law should not be read as the true asymptotic cost.

A position error ε translates into an angular error $\lesssim \varepsilon/\Delta_{\min}$, where $\Delta_{\min}$ is the minimum Earth–body distance (about 0.37 AU for Mars at a perihelic opposition). Angular accuracy requirements must therefore be set against $\Delta_{\min}$, not against a, and diverge for close approaches. The bound (16) is monotone in N; the realised error of a particular truncated set need not be at small N.

TABLE II. Near-parabolic limit: harmonics N certified by the bound (16) versus the minimal N for which the sampled maximum of $|z - z_N|/a$ over a fine M grid is at most tol (relative to a). *: extrapolated with $3CN^{-2/3}$ (approximate). At $e = 1$ the bound is infinite.

	tol $= 10^{-2}$		tol $= 10^{-6}$	
e	N_{cert}	N_{actual}	N_{cert}	N_{actual}
0.5	10	6	29	23
0.89	125	38	350	230
0.99	4,744	364	13,375	6,564
0.999	150,534	1,058	424,558	146,581
0.9999	4,761,856	1,327	13,430,703	2,852,676
0.999999	4.76×10^9	1,366	1.34×10^{10}	$1.22 \times 10^{9\,*}$
1	—	1,367	—	$1.37 \times 10^{9\,*}$

G. Perturbed motion: invariant tori, KAM and frequency analysis

Mutual perturbations break the exactness of the preceding subsections. Three cases should be kept apart. (i) *Exact Kepler expansions*: Props. 1 and 4 and Thm. 1 hold uniformly for all time. (ii) *Exact expansions on analytic invariant tori*: under the assumptions stated for Eq. (18) below, the representation is exact and uniformly convergent for all time [41]. (iii) *Empirical finite-window approximations* of trajectories not known to lie on such a torus, obtained by frequency analysis. Consider a near-integrable Hamiltonian $H = H_0(\mathbf{I}) + \mu H_1(\mathbf{I}, \boldsymbol{\theta})$ in action–angle variables.

Classical background. Lagrange–Laplace secular theory adds linear precession and node-regression frequencies g_i, s_i to the mean motions n_j, replacing the fixed (ω, Ω) of Subsec. IV C by slowly rotating frames [13, 29]. The Kolmogorov–Arnold–Moser (KAM) theorem states that, under non-degeneracy hypotheses on H_0 and for sufficiently small μ, a large measure of initial conditions with non-resonant (Diophantine) frequencies lies on *surviving* invariant tori carrying quasi-periodic motion [44–46]; the epicycle reading of this is due to Gallavotti [17]. KAM does not establish that an arbitrary bounded trajectory, or any particular one such as our Ceres integration, lies on such a torus. On an invariant torus with d rationally independent frequencies $\boldsymbol{\omega}$, assuming analytic dependence so that the embedding has absolutely summable Fourier coefficients,

$$\mathbf{r}(t) = \sum_{\mathbf{m} \in \mathbb{Z}^d} \mathbf{A}_{\mathbf{m}} e^{\mathrm{i}\langle \mathbf{m}, \boldsymbol{\omega} \rangle t}, \qquad \sum_{\mathbf{m}} |\mathbf{A}_{\mathbf{m}}| < \infty, \tag{18}$$

a Bohr almost-periodic function [8, 40, 41], so the expansion is exact and uniformly convergent for all t (case ii). Each term is a uniformly rotating vector, i.e. an epicycle, but the frequencies are integer combinations of $\boldsymbol{\omega}$

(for the Solar System, the n_j, g_i, s_i and a few resonance arguments [14, 41]), not multiples of one fundamental frequency. The geocentric vector subtracts two such expansions, so the two-family structure of Prop. 4 generalises: a union of the two quasi-periodic spectra, still without cross terms in the Cartesian vector.

Construction. The amplitudes and frequencies in Eq. (18) are obtained numerically from an integrated or ephemeris time series by Laskar's numerical analysis of fundamental frequencies (NAFF) [14–16, 40]: iteratively maximise the windowed Fourier amplitude over frequency, subtract the term found, and repeat, with Gram–Schmidt orthogonalisation of the basis (Sec. V). The analytic coefficients of the previous subsections give a prior against which the NAFF output can be validated.

Limits. Exactness for all time holds in cases (i) and (ii). For trajectories of case (iii), and for the use of case (ii) on the real Solar System, the following apply. The Ceres experiment of Sec. VI measures the reconstruction accuracy of a finite-window fit; it does not demonstrate that the motion is quasi-periodic.

1. *Chaos.* The inner planets are chaotic with a Lyapunov time of about 5 Myr [14]; a quasi-periodic representation then holds at best approximately on a finite window, with slowly drifting frequencies [42]. "Arbitrary precision for all time" is not available for such motion.

2. *Finite windows.* Over a record of length T the representation can be made accurate to the NAFF resolution; extrapolation outside the fitted window degrades with the drift rate.

3. *Frequency resolution.* Lines separated by less than $\sim 2\pi/T$ are in practice hard to separate: $2\pi/T$ is a practical resolution and conditioning scale for the method, not an impossibility threshold (a fit that imposes known frequencies can separate closer lines at the price of conditioning). Near-resonant combinations therefore need a longer record, and the number of terms needed grows with T.

4. *Near-resonances.* KAM tori are destroyed near resonances and the measure of surviving tori tends to one only as $\mu \to 0$; whether the actual planetary orbits lie on surviving tori is not implied by the theorem. For the planets $\mu \sim 10^{-3}$; the Jupiter–Saturn near-resonance (the Great Inequality) is a long-period term, not a breakdown, but it is the largest perturbation line.

5. *Unbound orbits.* For $e \geq 1$ the orbit is not bounded and $z(M)$ is not periodic; the series of Prop. 1 does not exist. Quasi-periodicity is a property of bounded motion. A hyperbolic flyby can be matched on any finite window by trigonometric polynomials (Weierstrass) but with no dynamical meaning or extrapolation. For $e \to 1^-$ the series converges; the order certified by Eq. (16) grows as $(1-e)^{-3/2}$, while the actual minimal order at fixed tolerance tends to a finite limit (Subsec. IV F).

6. *The Earth is not a Kepler point either.* Its orbital elements vary secularly, the Earth–Moon barycentre (not the Earth's centre) moves on the Kepler-like orbit, and the Sun is displaced from the barycentre by $\sim$ 0.005 AU mainly by Jupiter. These add frequency lines rather than conceptual obstacles.

V. NUMERICAL METHODS

All numerical results are produced by a small Python library (`numpy`, `scipy`) whose modules are named in Sec. V F. Units are astronomical units (AU) and days; angles are in radians; positions are heliocentric or geocentric ecliptic J2000 Cartesian vectors.

A. Kepler propagation

A body is specified by osculating elements $(a, e, i, \Omega, \omega, M_0)$ at an epoch, with $n = \sqrt{\mu/a^3}$, $\mu = k^2$ and $k = 0.01720209895$ rad/day the Gaussian gravitational constant. Kepler's equation $M = E - e\sin E$ is solved by Newton iteration on E with the Newton step clipped to $|\Delta E| \leq 1$, the mean anomaly first reduced to $[-\pi, \pi)$, and the starting value $E_0 = M + e\sin M$ for $e \leq 0.8$ and $E_0 = \pm\pi$ for larger e. Iteration stops when $|\Delta E| < 10^{-15}$ (or the residual is below 10^{-15}); the achieved residual $|E - e\sin E - M|$ is 10^{-15}–10^{-14}. The position is then $\mathsf{R}\,(a(\cos E - e),\, a s \sin E,\, 0)^{\mathsf{T}}$ with the rotation of Sec. IV. This direct solution is the "truth" against which the series of Prop. 1 is checked: for $e = 0.0167$, 0.1, 0.5 and $N = 60$ the maximum of $|z_N - z_{\mathrm{true}}|/a$ over 2×10^4 points in M is 9.6×10^{-16}, 1.0×10^{-15} and 1.06×10^{-14}, respectively, i.e. the coefficient formulas are verified to $\sim 10^{-14}$.

B. Analytic epicycle expansion

The coefficients c_k, $|k| \leq N$, are evaluated from the derivative-free forms Eqs. (4)–(5) with the Bessel functions $J_k(ke)$ and $J_{k+1}(ke)$ from `scipy.special.jv`. Two details matter for small e, where the factors s/e and $(1 \pm s)/e$ are singular term by term (Sec. IV A). First, $(1-s)/e$ is evaluated as $e/(1+s)$, avoiding the cancellation in $1-s$. Second, $J_k(ke)/e$ is formed from `jv` directly, which is relatively accurate for tiny e. The case $e = 0$ is handled exactly ($c_1 = 1$, all other $c_k = 0$). Each term $a c_k e^{\mathrm{i}kM}$ becomes one circle with radius $a|c_k|$, signed rate kn, phase $kM(t_{\mathrm{ref}}) + \arg c_k$ with $\arg c_k \in \{0, \pi\}$, lying in the orbital plane spanned by $\mathbf{P}, \mathbf{Q}$ with normal $\mathbf{N}$; the $k = 0$ term is a fixed arm of length $\frac{3}{2}ae$; terms with exactly zero radius are dropped.

The geocentric chain is the union of the body's circles and the Earth's circles with phase shifted by π (negation), all referenced to one common epoch, following Eq. (12); no cross terms are formed. The chain is evaluated by summing the circles. The truncation bound ε_N of Eq. (16), and the geocentric bound (17), are implemented directly from the formulas, with the special case $e = 0$ giving 0 for $N \geq 1$ and a for $N = 0$. Against the direct Kepler solution the truncated expansion with $N = 40$ for both body and Earth reproduces the geocentric positions of Mars, Eros and Ceres to 6×10^{-14}–8×10^{-13} AU over a -100 to $+300$ yr span, and the bound (16) was verified to hold for all tested N and e up to roundoff (10^{-13}).

C. Frequency analysis (NAFF)

For a perturbed trajectory the circles are not known analytically and are extracted from a uniformly sampled series $\mathbf{r}(t_n)$, $n = 0, \ldots, N_s - 1$, by the numerical analysis of fundamental frequencies of Laskar [14, 40]. We use the weighted inner product $\langle f, g \rangle = \sum_n w_n f_n \bar{g}_n$ with the Hann^p window $w_n \propto \sin^{2p}\big(\pi n/(N_s - 1)\big)$ normalised to $\sum_n w_n = 1$, and the default $p = 4$ (Laskar's $\sin^8$ window). The algorithm is as follows.

1. The constant (mean) vector is fitted first and removed.

2. The strongest peak of the windowed FFT of the current residual, summed over the three Cartesian components and zero-padded by a factor of four, is located.

3. The frequency is refined by maximising, with a bounded Brent search within ± 1 FFT bin of the peak, the power of the projection of the residual on the candidate basis $\{\cos \omega t, \sin \omega t\}$ *orthogonalised against all previously accepted basis functions* (Gram–Schmidt, implemented by QR/SVD in the weighted space). For real data the cosine and sine pair removes the interference between the $+\omega$ and $-\omega$ lines. Because Brent's tolerance is $\sqrt{\epsilon_{\mathrm{mach}}}$ relative to its argument, the search is carried out on the offset from the FFT peak in nested windows.

4. The projection is subtracted, and steps 2–3 are repeated for K terms or until the residual falls to 10^{-13} of the data norm; candidates whose Gram–Schmidt remainder is below 10^{-6} of their norm are rejected as indistinguishable from earlier ones.

5. The final amplitudes come from one joint weighted least-squares fit over all accepted frequencies and the constant, which spans the same subspace as the sequential amplitudes without error accumulation.

Each line $\omega > 0$ with complex vector amplitude $\mathbf{A}$ is the ellipse $\mathrm{Re}(\mathbf{A}e^{\mathrm{i}\omega t})$ and is converted to a prograde and a retrograde circle. Because a fitted $\mathbf{A}$ is a general complex vector, its ellipse need not have axes along known directions, so we project $\mathbf{A}$ onto an orthonormal basis $(\mathbf{u}, \mathbf{v})$ of the plane spanned by $\mathrm{Re}\,\mathbf{A}$ and $\mathrm{Im}\,\mathbf{A}$ and read the prograde and retrograde radii from $\alpha = (\mathbf{A}\cdot\mathbf{u} + \mathrm{i}\mathbf{A}\cdot\mathbf{v})/2$ and $\beta = (\mathbf{A}\mathbf{u} - \mathrm{i}\mathbf{A}\mathbf{v})/2$ as $|\alpha|, |\beta|$. This is equivalent to the singular value decomposition of $[\mathrm{Re}\,\mathbf{A}, -\mathrm{Im}\,\mathbf{A}]$, whose radii are $(\sigma_1 \pm \sigma_2)/2$ (Subsec. IV C); Prop. 3 is the Kepler special case.

Measured accuracy. Because the window vanishes at the record ends, the weighted fit is least accurate there, and we report both full-window and interior (middle 80% of the samples) errors below. On a 200-yr geocentric Mars series at 5-day sampling with 30 frequencies, the worst relative frequency and radius errors over the 10 largest analytic terms are 7×10^{-11} and 10^{-7} for $p = 4$ (with 10-day sampling and 24 frequencies, 2×10^{-10} and 9×10^{-8}, the pair quoted in Sec. VI D), 2×10^{-10} and 5×10^{-7} for $p = 3$, and $\sim 3 \times 10^{-9}$ and 5×10^{-4} or worse on the weak terms for $p = 2$; hence $p = 4$. Isolated well-separated lines on 100-period synthetic records are recovered to relative frequency accuracy 1.5×10^{-10}–4.6×10^{-10}, limited by the flatness of the objective near its maximum. On the 100-yr geocentric series of Eros the reconstruction error is 7.6×10^{-7} AU at worst (record ends) and 3.1×10^{-8} AU in the interior. The scale $2\pi/T$ (T the record length) is a practical resolution and conditioning scale, not a sharp impossibility threshold; with the Hann^p window the practical limit is roughly $(p + 1)$ bins before interference biases the frequencies, and closer pairs are returned as one merged, biased line (plus at most a poorly conditioned partner) without failure. We do not quote an asymptotic convergence exponent for the window.

D. N-body integration

The perturbed test case is a massless asteroid (Ceres) together with the Sun, the Earth–Moon barycentre, Jupiter and Saturn, integrated as Newtonian point masses in the *barycentric* frame with the Sun as a moving body (so no indirect terms are needed); heliocentric output is obtained by subtracting the Sun's barycentric position. The mass ratios $M_\odot/m$ are 328900.5614 (Earth+Moon), 1047.3486 (Jupiter system) and 3497.898 (Saturn system) (IAU 2012 / JPL DE430 values). Ceres, with $GM \approx 4.7 \times 10^{-10}\, M_\odot$, is treated as a test particle: it feels all massive bodies and exerts no force. The equations of motion are integrated with the adaptive eighth-order Dormand–Prince method (`scipy.integrate.solve_ivp`, `DOP853`) with relative tolerance 10^{-12} and absolute tolerance 10^{-14} (AU, AU/day), and dense output sampled at fixed intervals. Initial conditions are each body's osculating elements propagated by the Kepler solution of Sec. V A to the start epoch (planets: JPL approximate positions at J2000 with $\mu = k^2$; Ceres: JPL SBDB osculating elements), with the Sun placed so that the barycen-

tre is at rest. Heliocentric velocities come from two-body Kepler with $\mu = k^2$ rather than $k^2(1+m)$, which changes the osculating mean motions of the planets at the 10^{-3} (Jupiter) level; this is immaterial because the study compares against the same integration. With all planet masses set to zero the integration reproduces the Kepler propagation of Sec. V A to better than 10^{-8} AU over a 25-yr test span, a check of the integrator; this zero-perturber, 25-yr control is supplementary and does not by itself validate the 200-yr perturbed integration. The latter was therefore repeated with tighter tolerances, DOP853 with rtol $= 10^{-13}$ and atol $= 10^{-16}$ in place of 10^{-12} and 10^{-14}. Over the whole 200-yr sampled grid the geocentric Ceres positions of the two runs differ by at most 1.03×10^{-8} AU, dominated by the Earth (1.03×10^{-8} AU; heliocentric Ceres alone: 5.4×10^{-12} AU). This shows that the reference is insensitive to the tolerance at the level of 10^{-8} AU, three orders of magnitude below the smallest NAFF interior error reported below (1.7×10^{-5} AU); it is one tighter run, not a proof of convergence. The study uses a 200-yr span at 5-day sampling (14 611 samples) starting at the Ceres epoch, JD 2461200.5.

E. Baselines and error metrics

The NAFF reconstruction of the Ceres series is compared with Kepler baselines. In the main comparison (Sec. VI E) every reconstruction is fitted to the *heliocentric* integrated Ceres trajectory and the *same* integrated Earth–Moon barycentre trajectory is subtracted from it, so that the geocentric error contains no Earth-model error.

1. *Fitted six-element Kepler, unweighted:* a nonlinear least-squares fit of all six elements of Ceres (`scipy.optimize.least_squares`, started from the mean-a orbit, every fifth sample) to the heliocentric series.

2. *Fitted six-element Kepler, Hann^4-weighted:* the same fit with residuals weighted by $\sqrt{w_n}$, with w_n the NAFF window, so that both methods minimise the same weighted objective.

Three further baselines are reported as supplementary context. The *osculating Kepler* orbit uses the osculating elements at the start epoch (the full analytic chain with $N_b = N_\oplus \in \{10, 20, 40\}$); its error is dominated by the difference between osculating and mean elements and must not be read as a "Kepler versus epicycles" comparison. The *mean-a Kepler* orbit has the same elements but a replaced by the value corresponding to the dominant NAFF line of the heliocentric Ceres series, $a = (\mu/n_{\mathrm{mean}}^2)^{1/3}$. Finally, in the earlier protocol the NAFF chain and a six-element fit were both fitted to the *geocentric* series, with the Earth's Kepler orbit subtracted in the Kepler case, so that the Kepler baseline there cannot correct any Earth-model error. Reconstruction errors are Euclidean distances $d_n = |\mathbf{r}_n - \hat{\mathbf{r}}_n|$ in AU. We report the sup norm $\max_n d_n$ and the RMS $(\langle d_n^2 \rangle)^{1/2}$ over the full window, and the same two quantities over the *interior* 80% of the samples (indices $0.1N_s$ to $0.9N_s$), which excludes the window-edge effect described above. For the analytic bound (Table I) the error is the sampled maximum of $|z - z_N|/a$ over a grid in M.

F. Code availability and reproducibility

Archive and licence. The code, data products and this paper are in the repository `https://github.com/ebowman/epicycles`, archived as release v1.0. The interactive viewer is at `https://ebowman.github.io/epicycles/` and the paper PDF at `https://ebowman.github.io/epicycles/paper.pdf`. The code is released under the MIT licence; the paper and figures under CC BY 4.0.

Implementation. The implementation is in the repository's `epicycles` package: `kepler.py` (`Elements`, `solve_kepler`, `position`), `expansion.py` (`kepler_fourier`, `to_epicycles`, `geocentric_chain`, `evaluate_chain`), `bounds.py` (`rho`, `tail_bound`, `geocentric_tail_bound`), `naff.py` (`naff_complex`, `naff_vector`), `nbody.py` (`integrate`, `geocentric`) and `bodies.py` (element sets).

Initial conditions and epochs. Table III lists the element sets used (heliocentric ecliptic J2000; angles in degrees, a in AU), exactly as in `epicycles/bodies.py`. The planets are the JPL Solar System Dynamics "Approximate Positions of the Planets" Table 1 values at J2000 (JD 2451545.0), rates ignored, converted by $M_0 = L - \varpi$ and $\omega = \varpi - \Omega$; the Earth is the Earth–Moon barycentre. The asteroids are JPL Small-Body Database osculating elements at epoch JD 2461200.5 (TDB). Both sets were retrieved on 2026-10-14. Times are Julian dates (TDB); $\mu = k^2$ with $k = 0.01720209895$. The N-body study starts at JD 2461200.5, spans 200 yr (73 050 d) with 5-day output (14 611 samples), uses DOP853 with rtol $= 10^{-12}$, atol $= 10^{-14}$ and the mass ratios of Sec. V D. The geocentric figures of Sec. VI B use daily samples from JD 2461200.5 over 50 yr, and the Mars NAFF test uses 200 yr at 10-day sampling with $K = 24$ (Sec. VI D).

Software versions. Tested with Python 3.12.4 (the local interpreter), `numpy` 2.5.4, `scipy` 1.18.1 and `matplotlib` 3.11.2, as locked in `uv.lock`. The project supports Python $\geq$ 3.11 (`pyproject.toml`); the lock file resolves `scipy` 1.18.1 for Python $\geq$ 3.12 and 1.17.1 for Python 3.11 (and `numpy` 2.4.6 for Python 3.11); the Python 3.11 resolution was not run. The viewer uses three.js 0.186.1, TypeScript 5.9.3, Vite 8.3.1 and Vitest 5.0.2 on Node $\geq$ 22.12 (`web/.nvmrc`: Node 24).

Commands. From the repository root, after `uv sync`:

- `uv run pytest -q`: Python test suite.

TABLE III. Element sets used (angles in degrees, a in AU), as in `epicycles/bodies.py`. Planet ω and M_0 are derived from the JPL longitudes of perihelion and mean longitude.

Body	Epoch (JD)	a	e	i	Ω	ω	M_0
Earth	2451545.0	1.00000261	0.01671123	-1.531e-05	0	102.9376819	357.5268897
Mars	2451545.0	1.52371034	0.0933941	1.84969142	49.55953891	286.4968315	19.39019754
Jupiter	2451545.0	5.202887	0.04838624	1.30439695	100.4739091	274.2545707	19.66796068
Saturn	2451545.0	9.53667594	0.05386179	2.48599187	113.6624245	338.9364538	317.3553659
Ceres	2461200.5	2.765552595	0.07969229515	10.5880278	80.24862682	73.29421453	274.4193464
Eros	2461200.5	1.458243717	0.2228779628	10.8285441	304.2679713	178.9181319	62.51145502
Phaethon	2461200.5	1.271464621	0.8896722844	22.3105273	265.098806	322.3001685	301.4858239

- `uv run python experiments/perturbed_ceres.py` (≈ 3 min): Sec. VI E, Table VI and Fig. 6 data, the heliocentric baseline and the tolerance check (`experiments/out/perturbed_ceres.json`, `perturbed_ceres_tolerance.json`).
- `uv run python experiments/near_parabolic.py` (≈ 2–3 min): Table II (`near_parabolic.json`) and the NAFF line data of Table V (`naff_lines_mars.json`).
- `uv run python experiments/tables.py` (seconds): Tables I and IV (`experiments/out/nneeded.json`, `ptolemy_truncations.json`).
- `uv run python experiments/make_figures.py`: Figs. 1, 2, 3, 4, 6 and 5 (written to `figures/`; the screenshot of Fig. 7 is a manual capture of the viewer) and the data files for the viewer.
- `uv run python experiments/export_golden.py`: golden vectors for the TypeScript port.
- `npm --prefix web ci`, `npm --prefix web test`, `npm --prefix web run build`: viewer dependencies, tests and build.
- `make -C paper`: this paper.

The planar maximum and RMS of model A are also reproduced by `experiments/make_figures.py`.

VI. RESULTS

All numbers below are produced by `experiments/make_figures.py` from the code described in Sec. V; the planetary elements are approximate J2000 mean elements (Sec. VIII), so the results test the expansion and its truncation machinery against exact two-body motion and against an N-body integration, not against an ephemeris.

A. Coefficient decay

Figure 1 shows $|c_k|$ and $|c_{-k}|$ of the Kepler Fourier series (Prop. 1) for $e \in \{0.0934, 0.25, 0.6, 0.89\}$. The coefficients decay geometrically at the Kapteyn rate $\rho(e)^k$, $\rho = e\,e^s/(1+s)$, $s = \sqrt{1-e^2}$, with $\rho = 0.127$, 0.334, 0.742 and 0.964 respectively. Retrograde terms are smaller than prograde terms by roughly $(e/(1+s))^2$ at small k. The formulas themselves reproduce the exact Kepler solution to 10^{-14} (in units of a) at $e = 0.5$ with 60 harmonics.

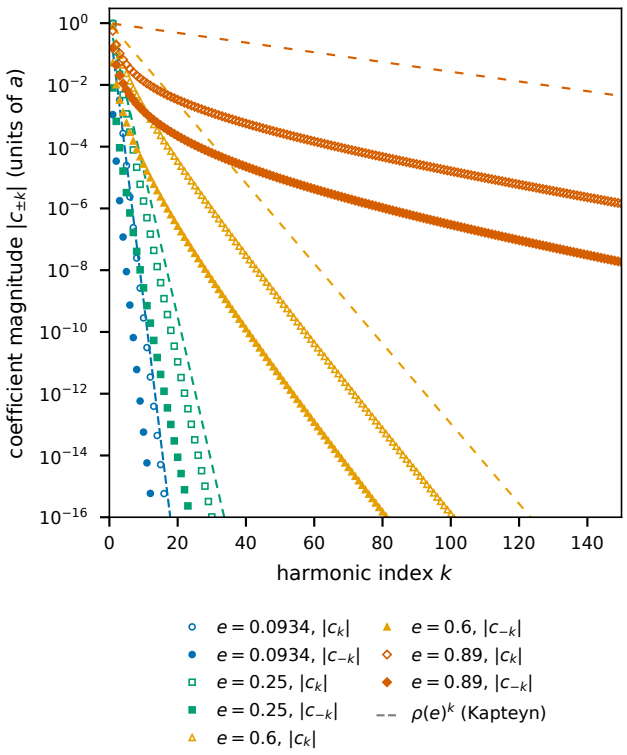


FIG. 1. Magnitude of the Kepler-orbit Fourier coefficients $|c_k|$ (open markers, $k > 0$, prograde) and $|c_{-k}|$ (filled markers, retrograde; marker shape distinguishes e) in units of a, for $e \in \{0.0934, 0.25, 0.6, 0.89\}$ (Mars, and three illustrative values; $e = 0.89$ is Phaethon). Dashed lines: the Kapteyn rate $\rho(e)^k$.

B. Geocentric convergence

Figure 2 shows the maximum geocentric error over a 50-yr window (daily samples) of the truncated chain with $N_{\rm body} = N_\oplus = N$, against exact Kepler motion, together with the rigorous tail bound $\varepsilon_N(\text{body}) + \varepsilon_N(\oplus)$. The bound is respected everywhere and the error saturates at the double-precision floor, $\sim 2 \times 10^{-13}$ AU. The smallest sampled N for which the error is below 10^{-6} AU is 6 for Mars, 6 for Ceres and 240 for Phaethon, and below 10^{-9} AU it is 10, 10 and 450. The cost is therefore $2N = 12$ to 20 rotating circles plus a static offset per orbit (24 to 40 rotating circles plus one combined offset in the geocentric chain) for planet-like eccentricities and of order a thousand for $e = 0.89$, consistent with the steep growth of the certified order $N_{\rm cert} \propto (1-e)^{-3/2}$ (not the asymptotic behaviour of the actual minimal order; Table II). For the single-body series the observed N for a sup error below $10^{-12}a$ is 12 for Mars and 575 for $e = 0.89$, and the analytic bound overestimates it by at most a factor of ≈ 1.5.

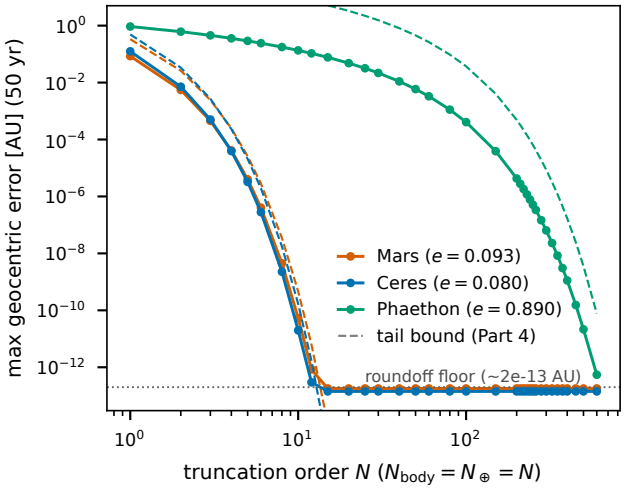


FIG. 2. Maximum geocentric position error [AU] over a 50-yr window (daily samples from JD 2461200.5) of the truncated epicycle chain with $N_{\rm body} = N_\oplus = N$ versus exact Kepler motion, for Mars, Ceres and Phaethon (solid), with the rigorous tail bound (dashed). Errors saturate at the roundoff floor ($\sim 2 \times 10^{-13}$ AU).

C. Low-order truncations of Mars

Table IV gives the longitude error of truncations of the geocentric Mars series in the planar approximation (coplanar J2000 mean elements, 60 yr at 0.5 d steps, exact two-body reference, no light time or aberration). The circular two-term approximation A, a single deferent and a single epicycle ($k = +1$ for both Mars and the Earth) with no eccentric offset and no equant, reproduces all 28 retrograde loops of the true motion, which is the qualitative achievement of the epicycle construction, but its longitude error reaches 33.2° (RMS 9.7°), at perihelic oppositions where the true distance is 0.37 AU against 0.52 AU in the circular model. Adding the $k = 0$ (eccentric) terms, model B, brings this to 9.4°, and the equant-level model C to 5.05°. None of these is a literal reproduction of Ptolemy's models, whose eccentric and equant agree with the $k = 0, 1, 2$ terms only to first order in e [9, 21]; in particular model A is not Ptolemy's model, and its 33.2° error is not an assessment of Ptolemy's accuracy. The table says what a harmonic truncation of the same order achieves. Truncation by retained set is not monotone at small orders (E is slightly worse than D, because the retrograde terms shift errors that partly cancelled), and a monotone decrease sets in only for $|k| \le 5$, where the error is 6×10^{-4} degrees. The $N = 5$ chain (10+10 rotating circles plus offsets) is within 0.001° over the window of Fig. 3.

TABLE IV. Low-order truncations of geocentric Mars (A is the circular two-term approximation, with no eccentric and no equant), planar approximation, 60 years. Longitude error against the exact two-body motion; the true number of retrograde intervals is 28 and every model reproduces it.

Model	Retained terms	max (°)	RMS (°)
A	$k = 1$, both bodies	33.2	9.7
B	$k \in \{0, 1\}$, both	9.40	2.7
C	Mars $k \in \{0, 1, 2\}$, Earth $k = 1$	5.05	0.97
D	$k \in \{0, 1, 2\}$, both	0.557	0.159
E	$\lvert k\rvert \le 2$, both	0.728	0.182
F	$\lvert k\rvert \le 5$, both	6×10^{-4}	1×10^{-4}
G	$\lvert k\rvert \le 10$, both	$< 10^{-4}$	$< 10^{-4}$

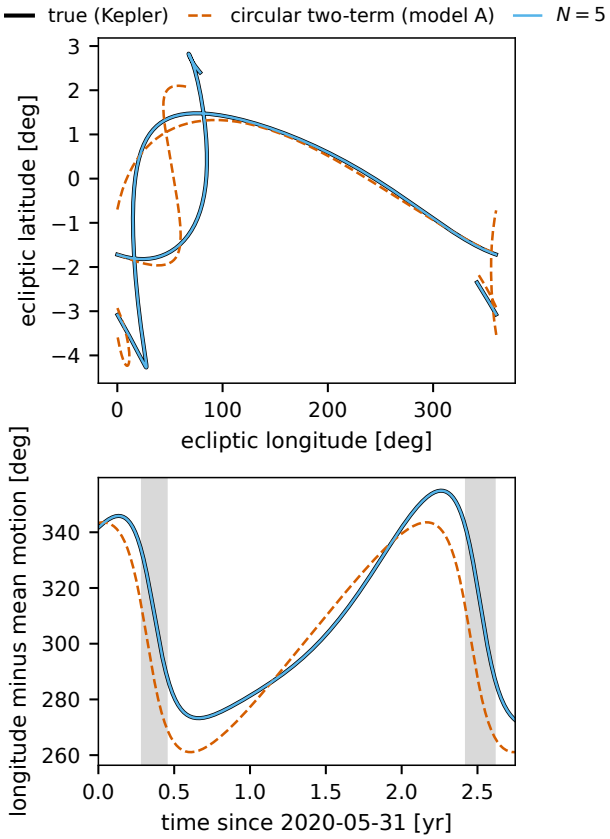


FIG. 3. Geocentric Mars over 2.75 years from 2020-05-31 (two oppositions). Top: sky track in ecliptic longitude and latitude. Bottom: unwrapped longitude minus its mean motion (0.524°/day); grey bands mark retrograde motion of the true path. Black: exact Kepler; dashed: circular two-term approximation (Mars $k = +1$ deferent plus Earth $k = +1$ epicycle only, with no eccentric or equant, so not Ptolemy's actual model; model A of Table IV); blue: $N = 5$ ($10 + 10$ rotating circles plus an offset each). The circular two-term approximation reproduces the retrograde loops but is displaced by up to 32.8° in longitude in this window (planar 60-yr sup norm 33.2°, RMS 9.7°); $N = 5$ is within 0.001°.

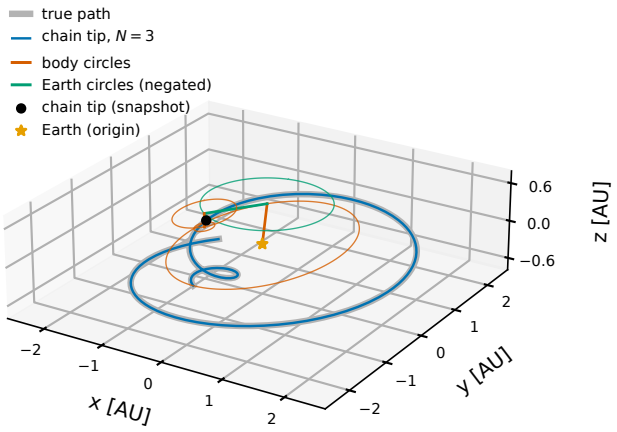


FIG. 4. Geocentric epicycle chain of Eros (inclination 10.8°) with $N = 3$ for Eros and for the Earth (12 rotating circles plus the static offsets, each circle drawn in its own orbital plane about the preceding arm tip; red: body circles, green: negated Earth circles). Blue: tip trail over one synodic period (845 d); grey: exact path. Maximum tip error 0.01 AU at this low order.

Figure 4 shows that inclination is handled without modification: each circle lies in its own orbital plane, and the chain for Eros ($i = 10.8°$) closes onto the exact path with the stated error.

D. NAFF on Keplerian trajectories

To validate the extraction we apply NAFF (Sec. V C, Hann[4] window) to the *exact* Keplerian geocentric Mars series over 200 yr sampled every 10 d with $K = 24$ lines, and match the lines to the analytic Bessel terms. For the ten largest analytic terms the worst relative frequency error is 2.0×10^{-10} and the worst relative radius error is 8.9×10^{-8}. On a 100-yr Eros series (5-day sampling, $K = 40$, $p = 4$) the reconstruction error is 7.6×10^{-7} AU at worst over the full window (record ends) and 3.1×10^{-8} AU over the interior 80%. The data-driven extraction therefore rediscovers the analytic epicycles; Table V lists the ten circles with their individual frequency, radius and phase errors.

E. Perturbed Ceres

The Ceres test integrates the Sun, Earth, Jupiter, Saturn and a massless Ceres over 200 yr (5-day sampling, Sec. V D). A NAFF chain fitted directly to the geocentric series would also have to absorb the perturbations of the Earth, which a Kepler fit of Ceres alone cannot do, and the NAFF window (Hann[4]) weights the centre

TABLE V. The ten largest analytic circles of the Keplerian geocentric Mars chain ($N_{\mathrm{body}} = N_{\oplus} = 40$, excluding the static offset) and their recovery by NAFF (200 yr, 10-day sampling, Hann^4 window, $K = 24$). Circles, not lines: the prograde and retrograde circles of one harmonic ($\pm k\, n_{\mathrm{M}}$) are recovered from a single NAFF line and so share the same frequency error. n_{M} and $n_{\oplus}$ are the mean motions of Mars and the Earth; the retrograde circles $-k\, n_{\mathrm{M}}$ are those with negative index. Errors are relative for frequency and radius and absolute (rad) for phase. Source: `experiments/out/naff_lines_mars.json`.

Circle	Radius (AU)	$\delta\omega/\omega$	$\delta r/r$	$\lvert\delta\phi\rvert$
$+1n_{\mathrm{M}}$	1.52	2.0×10^{-10}	8.9×10^{-8}	6.6×10^{-8}
$+1n_{\oplus}$	1.00	6.0×10^{-11}	6.5×10^{-8}	3.8×10^{-8}
$+2n_{\mathrm{M}}$	7.1×10^{-2}	2.1×10^{-11}	3.4×10^{-8}	1.4×10^{-8}
$+2n_{\oplus}$	8.4×10^{-3}	1.3×10^{-11}	1.6×10^{-14}	1.6×10^{-8}
$+3n_{\mathrm{M}}$	4.9×10^{-3}	7.2×10^{-11}	1.8×10^{-13}	7.2×10^{-8}
$-1n_{\mathrm{M}}$	1.7×10^{-3}	2.0×10^{-10}	8.9×10^{-8}	6.6×10^{-8}
$+4n_{\mathrm{M}}$	4.1×10^{-4}	1.2×10^{-14}	5.7×10^{-13}	7.5×10^{-12}
$+3n_{\oplus}$	1.0×10^{-4}	9.9×10^{-11}	2.6×10^{-12}	1.9×10^{-7}
$-2n_{\mathrm{M}}$	5.2×10^{-5}	2.1×10^{-11}	3.4×10^{-8}	1.4×10^{-8}
$+5n_{\mathrm{M}}$	3.7×10^{-5}	3.3×10^{-11}	2.0×10^{-11}	5.4×10^{-8}

of the record much more strongly than an unweighted least-squares fit does. To make the comparison like for like we therefore fit the *heliocentric* integrated Ceres trajectory in every case, and subtract the *same* integrated Earth trajectory from each reconstruction before measuring the geocentric error, so that Earth-model error is excluded from all rows. The Kepler baseline is a six-element least-squares fit, reported both unweighted and weighted by the same Hann^4 window as NAFF (residual weights $\sqrt{w}$). Errors are reported over the full window and over the interior 80% (Sec. V E).

Table VI gives the result. The interior error of the NAFF chain decreases overall with the number of lines K, from 2.6×10^{-2} AU at $K = 5$ to 1.7×10^{-5} AU at $K = 200$ (RMS 2.0×10^{-6} AU). The decrease is not monotone at the start: the interior maximum rises slightly from 2.6×10^{-2} AU at $K = 5$ to 2.9×10^{-2} AU at $K = 10$ and decreases monotonically for $K \geq 10$, while the interior RMS decreases at every step. The NAFF interior maximum is comparable to the fitted Kepler orbit (2.7×10^{-2} AU unweighted, RMS 1.2×10^{-2} AU) at $K =$ 5–10 and clearly smaller only for $K \gtrsim 50$ (1.1×10^{-2} AU at $K = 50$, 1.1×10^{-3} AU at $K = 100$). The full-window error does not fall: it lies between 2.2×10^{-2} and 4.4×10^{-2} AU for all K and grows with K beyond $K = 20$, because the window vanishes at the record ends and the extra lines fit the interior at the expense of the edges. The $K = 200$ chain is therefore worse in the maximum norm over the full window (4.4×10^{-2} AU) than either Kepler fit (3.7×10^{-2} AU unweighted, 2.9×10^{-2} AU weighted). Weighting the Kepler fit by the Hann^4 window does not help it in the interior: its interior maximum is slightly *worse* (2.8×10^{-2} against 2.7×10^{-2} AU) and only the full-window maximum improves. Fitting

TABLE VI. Perturbed Ceres, 200 yr: maximum geocentric error [AU] of reconstructions fitted to the *heliocentric* integrated Ceres trajectory, with the same integrated Earth subtracted from every row. NAFF with K lines ($2K$ rotating circles plus a static offset), and six-element Kepler fits unweighted and weighted by the Hann^4 window of NAFF. [a]Rotating circles, excluding the static offset.

	circles[a]	interior	full
NAFF $K = 5$	10	2.6×10^{-2}	2.6×10^{-2}
NAFF $K = 20$	40	1.9×10^{-2}	2.2×10^{-2}
NAFF $K = 50$	100	1.1×10^{-2}	3.0×10^{-2}
NAFF $K = 100$	200	1.1×10^{-3}	3.5×10^{-2}
NAFF $K = 200$	400	1.7×10^{-5}	4.4×10^{-2}
Kepler fit, unweighted	1 orbit	2.7×10^{-2}	3.7×10^{-2}
Kepler fit, Hann^4-wtd	1 orbit	2.8×10^{-2}	2.9×10^{-2}

NAFF to the geocentric series directly (the earlier protocol, Fig. 6) gives interior errors larger by a factor of about 3 at $K = 200$ (5.4×10^{-5} against 1.7×10^{-5} AU), and moving the Kepler baseline from the geocentric to the heliocentric protocol changes its interior maximum only from 2.74×10^{-2} to 2.67×10^{-2} AU. The comparison therefore does not hinge on the Earth model, and we do not attribute the NAFF improvement to the representation of Ceres alone: part of the difference between the two NAFF protocols is the additional Earth lines that a geocentric fit must carry. Osculating elements at the epoch give a much larger error, 0.89 AU, an artifact of osculating versus mean elements that we do not regard as a meaningful comparison.

Figure 5 shows why the full-window and interior numbers differ. The NAFF residuals are smallest at the centre of the record, rise gradually towards both ends, and climb steeply within the first and last 10% to the size of the Kepler residuals, where the Hann^4 weight is negligible and the fit is essentially unconstrained. This is the endpoint effect, and it worsens as lines are added because they are placed to reduce the weighted, mostly interior, residual. The Kepler residuals, weighted or not, oscillate at the 10^{-2} AU level across the whole window, being limited by the perturbation terms that a single ellipse cannot represent; the weighting reshapes them only slightly.

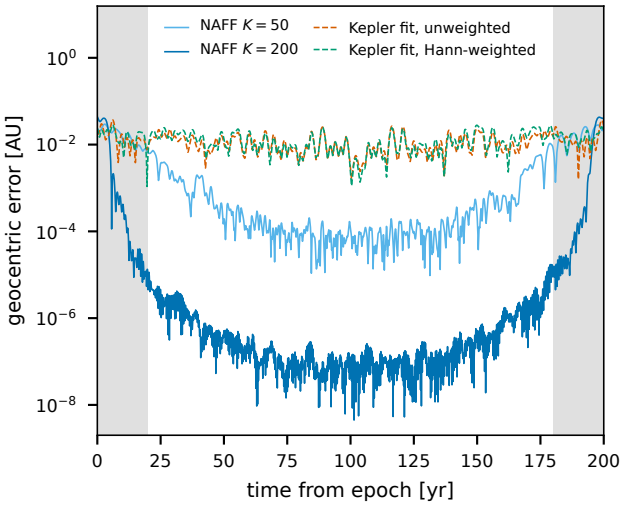


FIG. 5. Geocentric position error [AU] of reconstructed Ceres versus time over the 200-yr N-body window, with the same integrated Earth subtracted from every reconstruction. NAFF chains ($K = 50$, $K = 200$) are fitted to heliocentric Ceres with the Hann[4] window; the six-element Kepler fits to heliocentric Ceres are shown unweighted and weighted by the same window. Grey bands mark the first and last 10% of the record, outside the interior 80% used for the interior statistics. The NAFF errors rise towards the window ends, where the window weight vanishes.

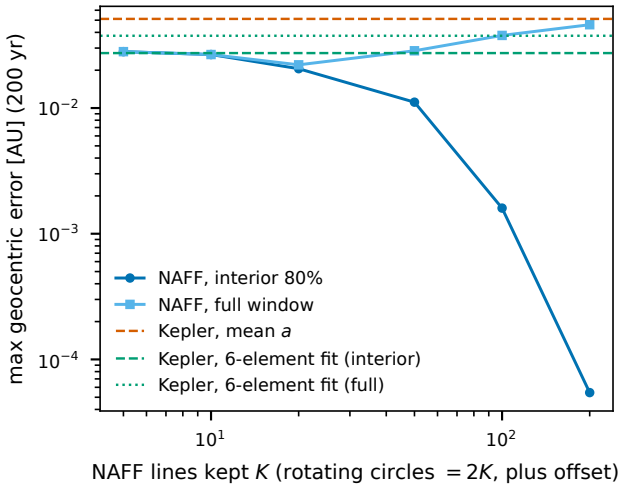


FIG. 6. Maximum geocentric position error [AU] of a NAFF-extracted epicycle chain ($2K$ rotating circles plus a static offset) fitted directly to the N-body (Sun, Earth, Jupiter, Saturn) geocentric Ceres trajectory over 200 years (so Earth's perturbations are fitted too), versus the number of lines K: interior 80% of the window (dark blue) and full window (light blue). Horizontal lines: Kepler baselines that fit only Ceres with the Earth held on a Kepler orbit, mean-a Kepler 0.051 AU and 6-element least-squares Kepler fit 0.027 AU (interior), 0.038 AU (full). The osculating-element Kepler orbit (error 0.89 AU) is omitted as an element-offset artifact. The matched comparison is Table VI.

VII. VISUALIZATION

To make the construction tangible we provide an open-source interactive viewer, a single-page web application (TypeScript and three.js) in the `web/` directory of the repository. Figure 7 shows it. Two three-dimensional views are synchronized to one clock. The left view is heliocentric and Newtonian: the Sun, the Earth and the chosen body move on their Kepler ellipses. The right view is geocentric, centred on the Earth, and shows the epicycle chain: the deferent, the successive epicycles drawn as circles in their own orbital planes, the arm joining their centres, and the trail of the chain tip. In the left panel a line joins the Earth to the body; it is the geocentric vector whose epicycle sum is drawn in the right panel.

The geocentric chain is computed live in the browser, by a TypeScript port of the Kepler and Bessel-coefficient code of Sec. V B that is tested against golden values generated by the Python reference. Nothing is precomputed or fitted: the radii, planes and angular rates are the analytic quantities of Sec. IV, determined by the orbital elements. The slider N (1 to 60) sets the truncation order for both the body and the Earth, so the viewer demonstrates the convergence of Sec. VI B directly: at $N = 1$ one sees the Ptolemaic deferent-and-epicycle picture with its visible displacement, and as N increases the chain tip

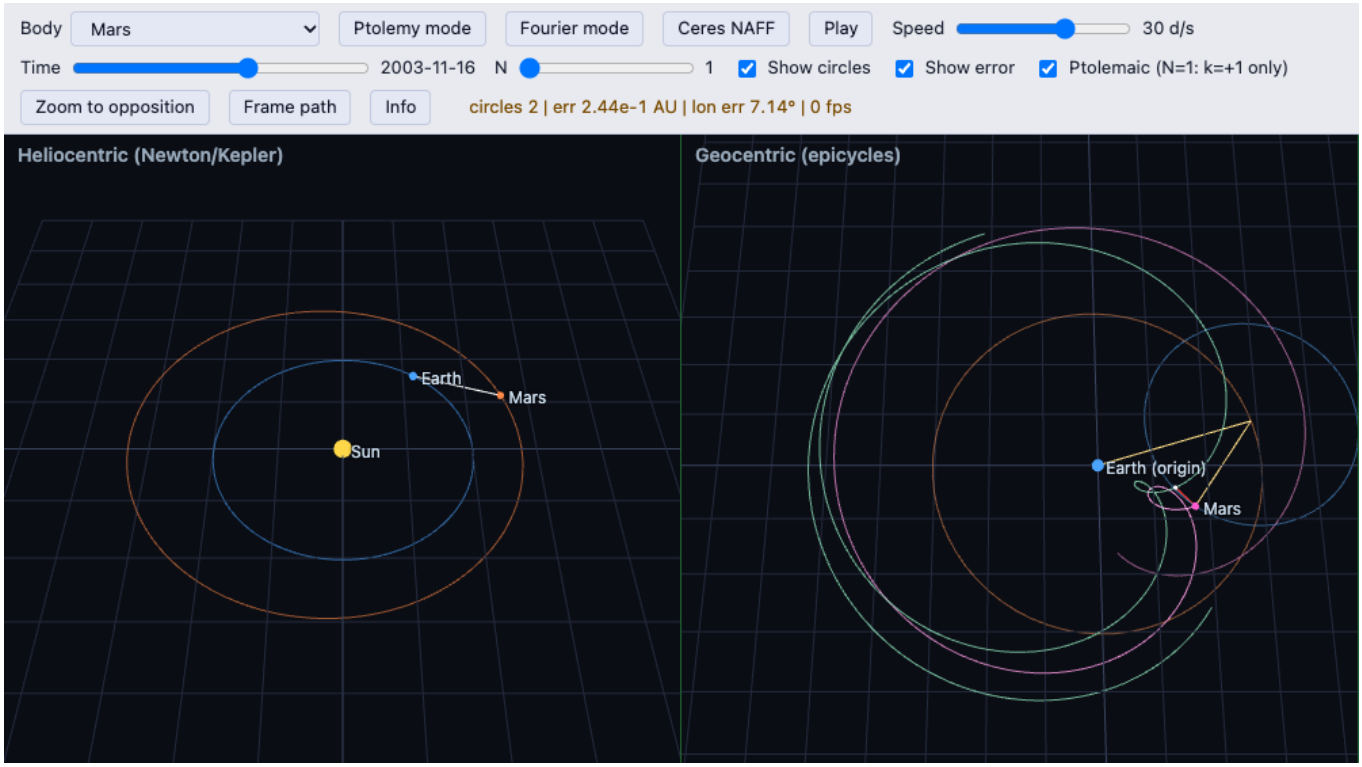


FIG. 7. Screenshot of the interactive viewer (URL parameters `preset=ptolemy&t=2452960&N=1&info=0`: Mars, Julian date 2452960, Ptolemaic $k = +1$ subset only, error shown). Left: heliocentric Newtonian view. Right: geocentric view with the epicycle chain (yellow arm), its two circles, the trail of the chain tip, the exact geocentric path (green) and the error vector (red) between the chain tip and true Mars. The readout (top) shows the circle count and the error against exact Kepler motion (here 0.244 AU, 7.14° in longitude).

converges onto the heliocentric truth and the small circles shrink. A live readout shows the number of circles and the instantaneous error against exact Kepler motion, in AU and as a longitude error. The state is encoded in URL parameters (`t`, a Julian date, `N`, `speed` in days per second, and `paused`) so that a configuration can be shared. Intended uses are as a classroom demonstration of what "epicycles are Fourier series" means quantitatively, and of how fast the series converges for planet-like eccentricities.

VIII. DISCUSSION

In what sense the hypothesis holds. The statement "planetary motion is a system of epicycles" is true in four distinct and increasingly weak senses. (i) For an unperturbed bound Kepler orbit the heliocentric position is *exactly* a Fourier series in the mean anomaly, uniformly convergent for every $e < 1$, and each term, a vector in an orbital plane, is the sum of two counter-rotating circles (Props. 1 and 3). (ii) The geocentric position is the sum of two such series, one for the body and one for the Earth, so it is an epicycle system with radii and rates fixed by the orbital elements. It is, however, not a Fourier series in the strict sense: the two mean motions are in general incommensurate, so the function is Bohr almost-periodic, with frequency module $\mathbb{Z}n_b + \mathbb{Z}n_\oplus$. There are no cross terms at combination frequencies in the Cartesian geocentric vector, because the difference is linear; numerically the amplitudes at combination frequencies are 10^{-16} to 10^{-15} relative, with a single line at 8.4×10^{-11}, and the out-of-mask $\sin^8$ window leakage is 3×10^{-8}. (iii) On an analytic invariant torus of a near-integrable system, under the assumptions of Eq. (18), the position is again exactly and uniformly a (multi-frequency) epicycle sum for all time; KAM theory supplies such tori only under additional hypotheses and for a subset of initial conditions, and does not show that a given body lies on one. (iv) For any other trajectory, such as our N-body Ceres integration, a finite set of epicycles recovered by frequency analysis is an empirical approximation on a finite window. The Ceres experiment demonstrates reconstruction accuracy there, with interior errors that decrease with the number of lines (monotonically for $K \geq 10$), and shows its price, namely window-edge errors that do not; it does not demonstrate that the motion is quasi-periodic.

The representation is exact only in cases (i)–(iii); in the cases flagged in the theory it fails, or is only a finite-window approximation. For unbound orbits ($e \geq 1$) the series does not exist. For chaotic motion the frequencies drift and a fixed set of circles is valid only on a finite time.

Over long windows secular drift of the elements limits any fixed chain, and in practice the resolution of two lines is set by a scale of about $2\pi/T$ for a record of length T (a conditioning scale, not an absolute limit). High eccentricity is also expensive: about 450 harmonics per orbit are needed for 10^{-9} AU at $e = 0.89$ (sampled). The order certified by the bound grows as $(1-e)^{-3/2}$, but this is not the asymptotic behaviour of the actual minimal order, which tends to a finite limit at fixed tolerance as $e \to 1$ (Table II). Finally, the Cartesian bound does not bound the angular error near close approaches, where it scales as $1/\Delta$.

"Epicycles can fit anything." Hanson's observation [4] and its philosophical consequence [10] are correct for an arbitrary curve, and a data-driven frequency analysis is, in that sense, a fit; our NAFF chains are fitted. For the analytic construction the objection does not apply: every radius and rate follows from $(a, e, i, \omega, \Omega, M_0)$ of the two orbits, and nothing is adjusted to the geocentric data. The number of circles is a truncation parameter with a rigorous error bound, not a freedom to absorb discrepancies. This answers the objection only for the dynamically derived expansion, not for Ptolemy's own free parameters, and the NAFF case is a measurement of the spectrum of a trajectory rather than an independent test.

Relation to prior work. As Sec. III describes, the conceptual core is Gallavotti's [17], the Bessel expansion is classical [11, 18], and the numerical FFT route is that of Peralta et al. [35]; modern ephemerides are likewise sums of uniform circular motions [19, 20]. Our results are a quantitative, explicitly geocentric and three-dimensional assembly of these ideas, with truncation bounds and a check of NAFF against the analytic terms.

Limitations. (1) We do not validate against JPL Horizons or any observed ephemeris; the reference for Kepler tests is exact two-body motion and for Ceres an N-body integration with only the Sun, the Earth, Jupiter and Saturn. (2) The planetary elements are approximate J2000 mean elements and no light time, aberration, precession or nutation is modelled, which changes the numbers but not the structure of the argument. (3) The models in Table IV, including the circular two-term approximation A, are harmonic truncations of the Kepler series, not reconstructions of Ptolemy's models. (4) The NAFF edge effects are significant: over the full window the extracted chain is no better than a fitted Kepler orbit, and our tolerance check on the numerical reference is a single tighter rerun, which shows insensitivity to the tolerance rather than proving convergence. (5) Truncation error is not monotone in the retained set for small orders. (6) Several historical sources were consulted only through secondary accounts (Sec. II).

IX. CONCLUSION

For a bound Newtonian two-body orbit, the geocentric position of a body is exactly a sum of two Bessel–Fourier epicycle families whose radii, planes and rates are determined by the orbital elements. We gave the explicit three-dimensional expansion with truncation bounds and verified them: $N \approx 6$ to 10 harmonics, i.e. $2N = 12$ to 20 rotating circles plus an offset per orbit, reach 10^{-6} to 10^{-9} AU for Mars and Ceres, hundreds are needed at $e = 0.89$, and a circular two-term approximation of Mars (not Ptolemy's actual model) reproduces the retrograde loops but is off by up to 33°. Frequency analysis recovers the analytic epicycles from Keplerian data to $\sim 10^{-10}$ in frequency and, for a perturbed Ceres trajectory, represents the interior of a 200-yr window with errors below 10^{-4} AU at $K = 200$ while remaining edge-limited over the whole window, where it is no better than a Kepler ellipse fitted to the same heliocentric data with matched weights. The right statement of the hypothesis is therefore bounded and finite-window: exact for Kepler motion, almost-periodic for the geocentric view, exact on analytic invariant tori, an empirical finite-window approximation for other perturbed motion, and absent for unbound orbits. The accompanying interactive viewer makes the convergence visible. Natural next steps are validation against a modern ephemeris, a treatment of edge effects by better windows or extrapolation, and extension to more bodies.

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Appendix A: Bessel and Kapteyn identities

We collect the classical identities used in Sec. IV [11, 12].

Generating function and Bessel's integral. For integer n,

$$e^{\mathrm{i}x\sin\tau} = \sum_{n\in\mathbb{Z}} J_n(x) e^{\mathrm{i}n\tau},$$
$$J_n(x) = \frac{1}{2\pi}\int_0^{2\pi} e^{\mathrm{i}(n\tau - x\sin\tau)}\, d\tau, \tag{A1}$$

with $J_{-n}(x) = (-1)^n J_n(x) = J_n(-x)$; J_n is real for real x, and the integral is unchanged under $\tau \to -\tau$ combined with $n \to -n$ so that the sign of the exponent in Lemma 1

is immaterial. The cosine and sine forms are

$$J_n'(x) = \frac{1}{\pi}\int_0^\pi \sin\tau\,\sin(n\tau - x\sin\tau)\,d\tau,$$
$$\tfrac{1}{2}\big[J_{n-1}(x)+J_{n+1}(x)\big] = \frac{1}{\pi}\int_0^\pi \cos\tau\cos(n\tau - x\sin\tau)\,d\tau. \tag{A2}$$

Recurrences. Differentiating Eq. (A1) in x and in τ gives

$$J_{n-1}(x)+J_{n+1}(x) = \frac{2n}{x}J_n(x),$$
$$J_{n-1}(x)-J_{n+1}(x) = 2J_n'(x), \tag{A3}$$

and, adding and subtracting, $J_n'(x) = \frac{n}{x}J_n(x) - J_{n+1}(x)$. At $x = ke$ with $n = k$ the first relation reads $J_{k-1}(ke) + J_{k+1}(ke) = \frac{2}{e}J_k(ke)$, which converts the first form of Eq. (4) into the derivative-free second form, and, with $J_k'(ke) = J_k(ke)/e - J_{k+1}(ke)$, gives Eqs. (6).

Proof of Lemma 1. Since $e^{-\mathrm{i}kM}dM = d(e^{-\mathrm{i}kM})/(-\mathrm{i}k)$, integration by parts gives

$$\frac{1}{2\pi}\int_0^{2\pi} e^{\mathrm{i}mE}e^{-\mathrm{i}kM}dM$$
$$= \frac{m}{k}\frac{1}{2\pi}\int_0^{2\pi} e^{-\mathrm{i}[(k-m)E - ke\sin E]}\,dE = \frac{m}{k}\,J_{k-m}(ke), \tag{A4}$$

using $M = E - e\sin E$ and Eq. (A1). The boundary term vanishes because $e^{\mathrm{i}mE}e^{-\mathrm{i}kM}$ is 2π-periodic in M.

Kapteyn's inequality. For $\nu \geq 0$ and $0 < \zeta \leq 1$,

$$|J_\nu(\nu\zeta)| \leq \left(\frac{\zeta\, e^{\sqrt{1-\zeta^2}}}{1+\sqrt{1-\zeta^2}}\right)^\nu \tag{A5}$$

is due to Kapteyn [18]; see Watson [12], §8.7 for the proof, which is not reproduced here. (We also verified the inequality numerically for $1 \leq k \leq 399$ and $e \in \{0.0167, 0.0934, 0.25, 0.6, 0.89, 0.99\}$.) It is the large-order, uniform-in-argument form of the Debye asymptotics $J_\nu(\nu\,\mathrm{sech}\,\alpha) \sim e^{\nu(\tanh\alpha - \alpha)}/\sqrt{2\pi\nu\tanh\alpha}$ with $\zeta = \mathrm{sech}\,\alpha$, which shows that the geometric rate ρ^k in Eq. (8) is sharp up to a factor of order $k^{-1/2}$ for fixed e.

Small-e limits of c_k. From the series $J_n(x) = \sum_{m\geq 0}\frac{(-1)^m}{m!\,(m+n)!}(x/2)^{2m+n}$ one has $J_k(ke) = \frac{(ke/2)^k}{k!}\big[1 - \frac{(ke/2)^2}{k+1} + \dots\big]$, so $J_k(ke) = O(e^k)$ and $J_{k+1}(ke) = O(e^{k+1})$. With $(1+s)/e = 2/e - e/2 + \dots$ and $(1-s)/e = e/2 + \dots$ in Eqs. (4)–(5):

$$c_k = \frac{k^{k-1}}{2^{k-1}\,k!}\,e^{k-1}\big[1 + O(e^2)\big],$$
$$c_{-k} = O(e^{k+1}), \qquad k \geq 1. \tag{A6}$$

In particular $c_1 \to 1$, $c_2 \sim e/2$, $c_3 \sim 3e^2/8$, and $c_{-1} \sim e^2/8$, $c_{-2} \sim e^3/24$ (Eq. (7)), and every coefficient is finite as $e \to 0$ although individual factors $1/e$ in Eq. (4) are not. Retrograde circles are thus smaller than the corresponding prograde ones by a factor $O(e^2)$ at small k (e.g. $|c_{-1}/c_1| \sim e^2/8$).

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